

DRYING

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Introduction

- ❖ Drying is the removal of a liquid from a material by the application of heat and is accomplished by the transfer a liquid from a surface into an unsaturated vapor phase.
- ❖ *Drying involves both heat and mass transfer*
- ❖ Drying is commonly the last stage in a manufacture process.
- ❖ Non –thermal drying
 - ✓ Expression of a solid to remove liquid
 - ✓ Adsorption by desiccant (desiccation)
 - ✓ Extraction of liquid from a solid by the use of solvent

Equipments for Drying

- ❖ Various equipments for drying are available and the choice of which depends on the f/g factors:
 - *Heat sensitivity of the product*
 - *Physical characteristics of the product*
 - *Nature of solvent to be removed*
 - *Aseptic condition requirement*
 - *Quantity of product to be dried*
 - *Source of heat and cost involve*

Equipments for Drying...

❖ A number of methods have been suggested for the classification of driers:

❖ **Based on physical characteristics**

1. Driers for dilute suspensions

- *Drum driers*
- *Spray driers*

2. Driers for solid materials

- *Tray drier*
- *Tunnel drier*
- *Rotary drier*
- *Fluidized bed drier*
- *Vacuum drier*
- *Freeze drier*

Equipments for Drying...

❖ Based on heat transfer methods

1. Convection dryers

- Tray/shelf
- Tunnel
- Rotary
- Fluidized bed

2. Conduction dryers

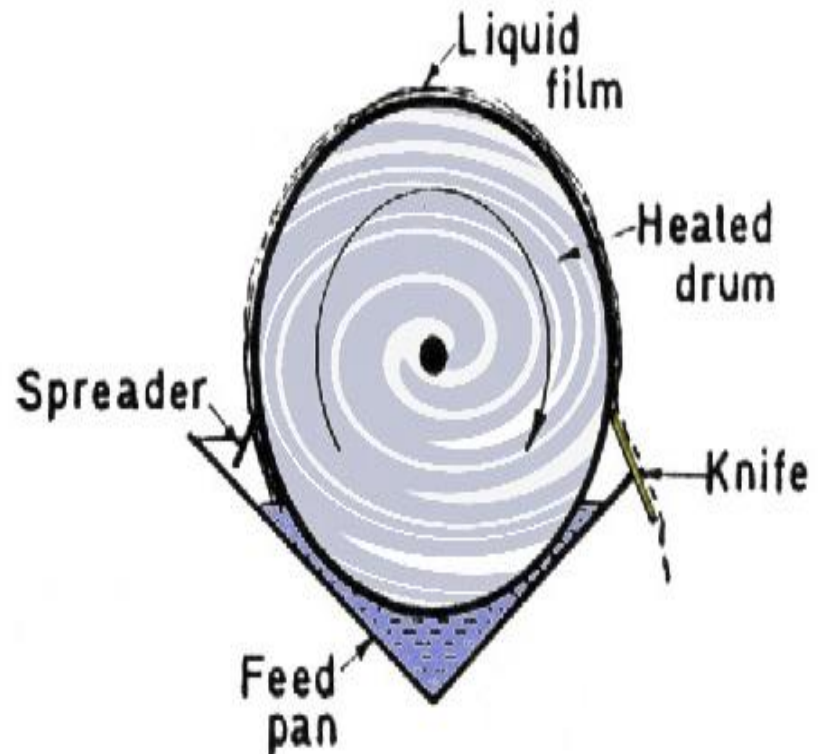
- Vacuum
- Freeze dryer

3. Radiation dryers

- drum

Drum Dryer

- ❖ It consists of a drum of about **0.75 - 1.5 m in diameter** and **2-4m in length**, heated internally, usually by steam, and rotated on its longitudinal axis.
- ❖ Liquids to be dried is kept in feeding pan
- ❖ Rotation of drum causes the liquid to spread as a **film** which dries rapidly
- ❖ Dried product is scrapped from the surface of the drum with the help of **knife**



Drum Dryer...

Advantages

- ❖ The method gives rapid drying, the thin film spread over a large area resulting in rapid heat and mass transfer.
- ❖ The equipment is compact, occupying much less space than other dryers.
- ❖ The drum can be enclosed in a vacuum jacket, enabling the temperature of drying to be reduced.
- ❖ It can handle a variety of materials

e.g. starch products, ferrous salts

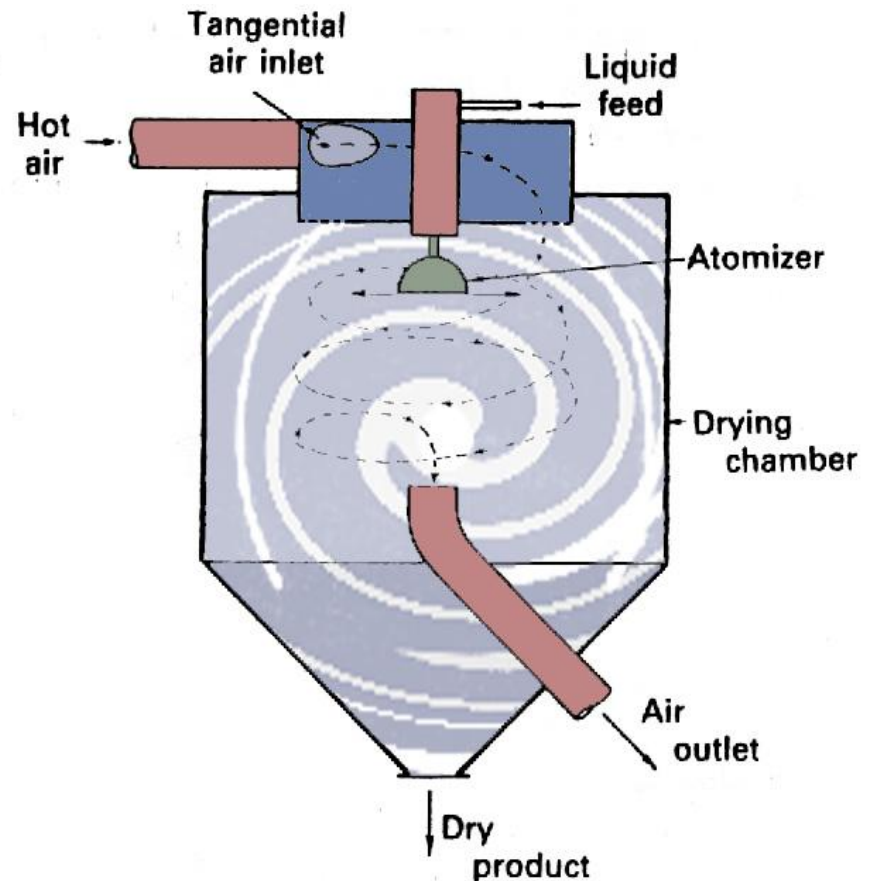
Drum Dryer...

Disadvantages

- ❖ The only disadvantage is that operating conditions are critical and it is necessary to introduce careful control on:
 - *feed rate,*
 - *film thickness,*
 - *speed of drum rotation and*
 - *drum temperature.*

Spray Dryer

- ❖ Here the liquid is dispersed as ***fine droplets*** into a moving stream of hot air by **atomizer** where they are evaporated rapidly before reaching the wall of chamber
- ❖ Product dries as fine powders which is collected
- ❖ Rate of feed is adjusted so as to dry each drop of sprayed liquid



Spray Dryer...

Advantages

- ❖ Drying rate is fast
- ❖ Thermolabile substances can be easily dried as rate of evaporation is high
- ❖ Dried powder has a uniform size and shape, good flow properties

❖ Disadvantages

- ❖ The equipment is bulky and costly
- ❖ Lot of wastage of heat

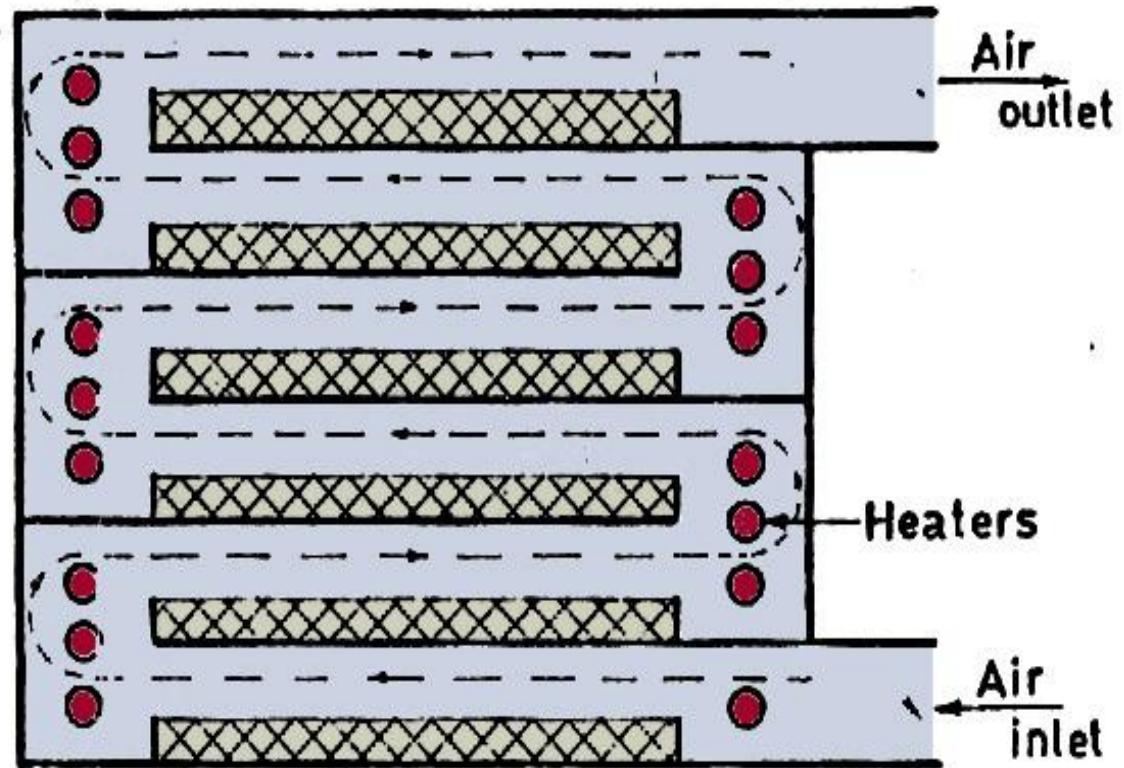
Spray Dryer...

❖ Substances which can be spray dried include:

- *Boric acid*
- *Methyl cellulose*
- *Lactose*
- *Citric acid*
- *Acacia*
- *Starch*

Tray Dryer

- ❖ Material to be dried is spread in thin layers in trays
- ❖ Heating is accomplished by means of hot air circulated over the trays
- ❖ Uses include drying of:
 - *crude drug,*
 - *chemicals,*
 - *powders,*
 - *tablet granules*
 - *items of equipments*



Tunnel Dryer

- ❖ Here the drying methods resembles the tray dryer but the oven is being replaced by a tunnel
- ❖ Material to be dried is entered at one end so that the outgoing material met the incoming air to ensure maximum drying and the dried material is collected at the other end of the tunnel
- ❖ Trays containing the wet material are loaded on single or multiple conveyer
- ❖ Applications are similar with tray dryers but more suitable for *large scale production*

Tunnel Dryer...

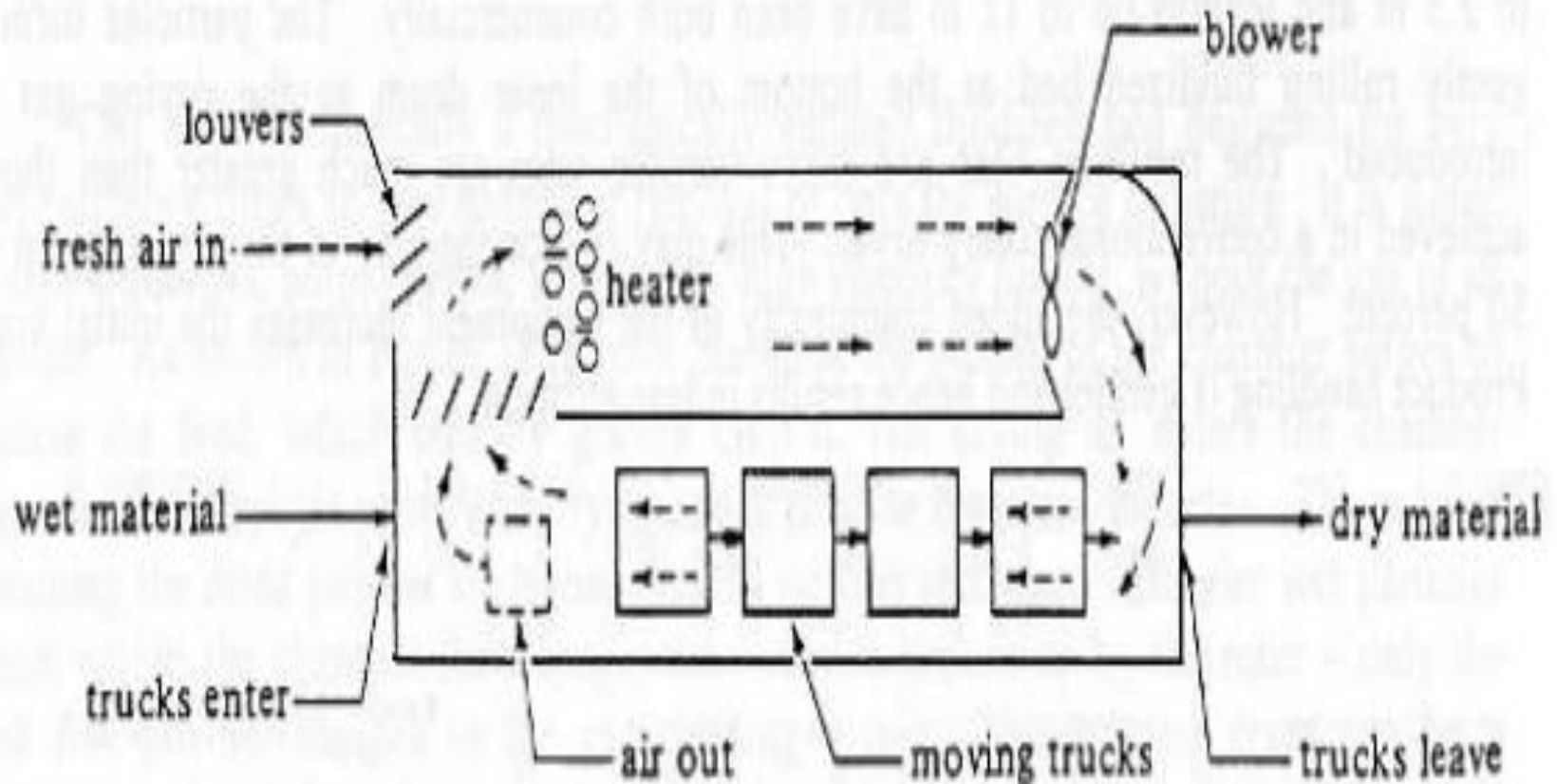
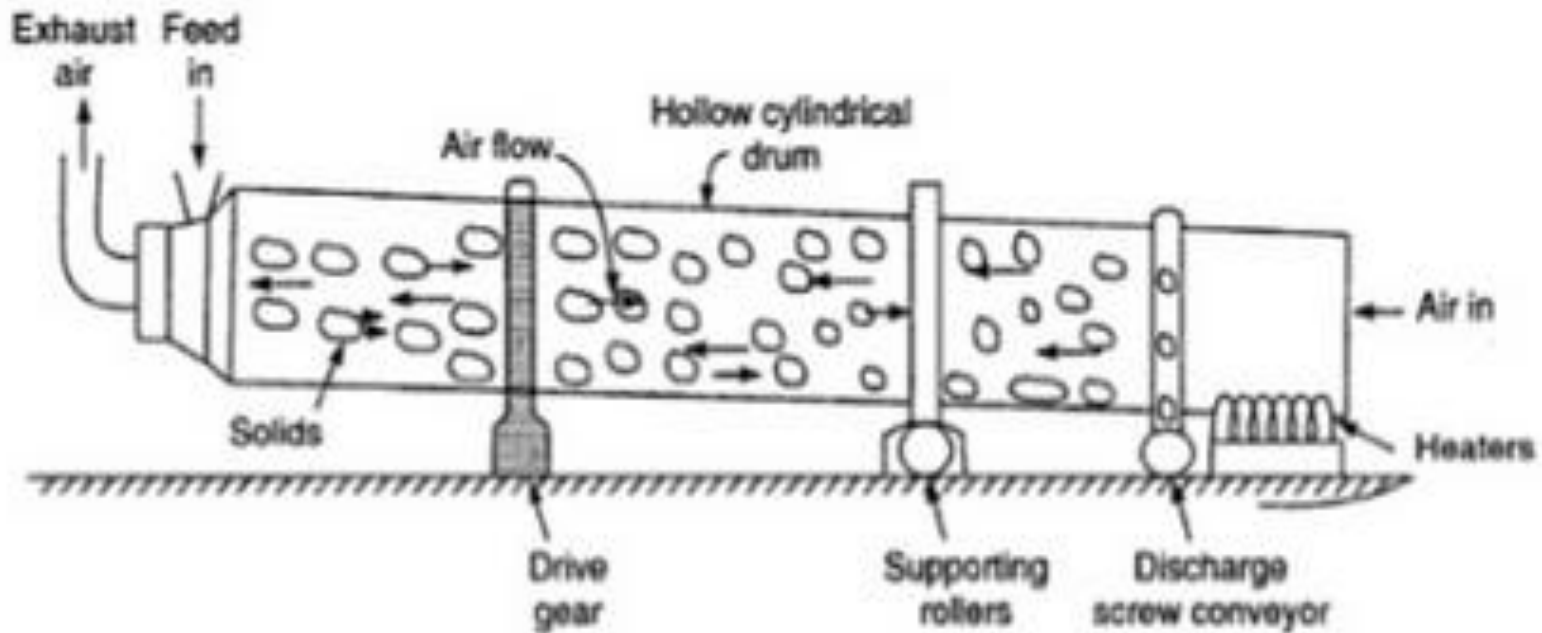


Fig: tunnel dryer

Rotary Dryer

- ❖ It is a modified form of tunnel dryers which consist of horizontal tube, slightly inclined, so that the material fed at one end will move gradually through the tube towards the other end as the tube rotates



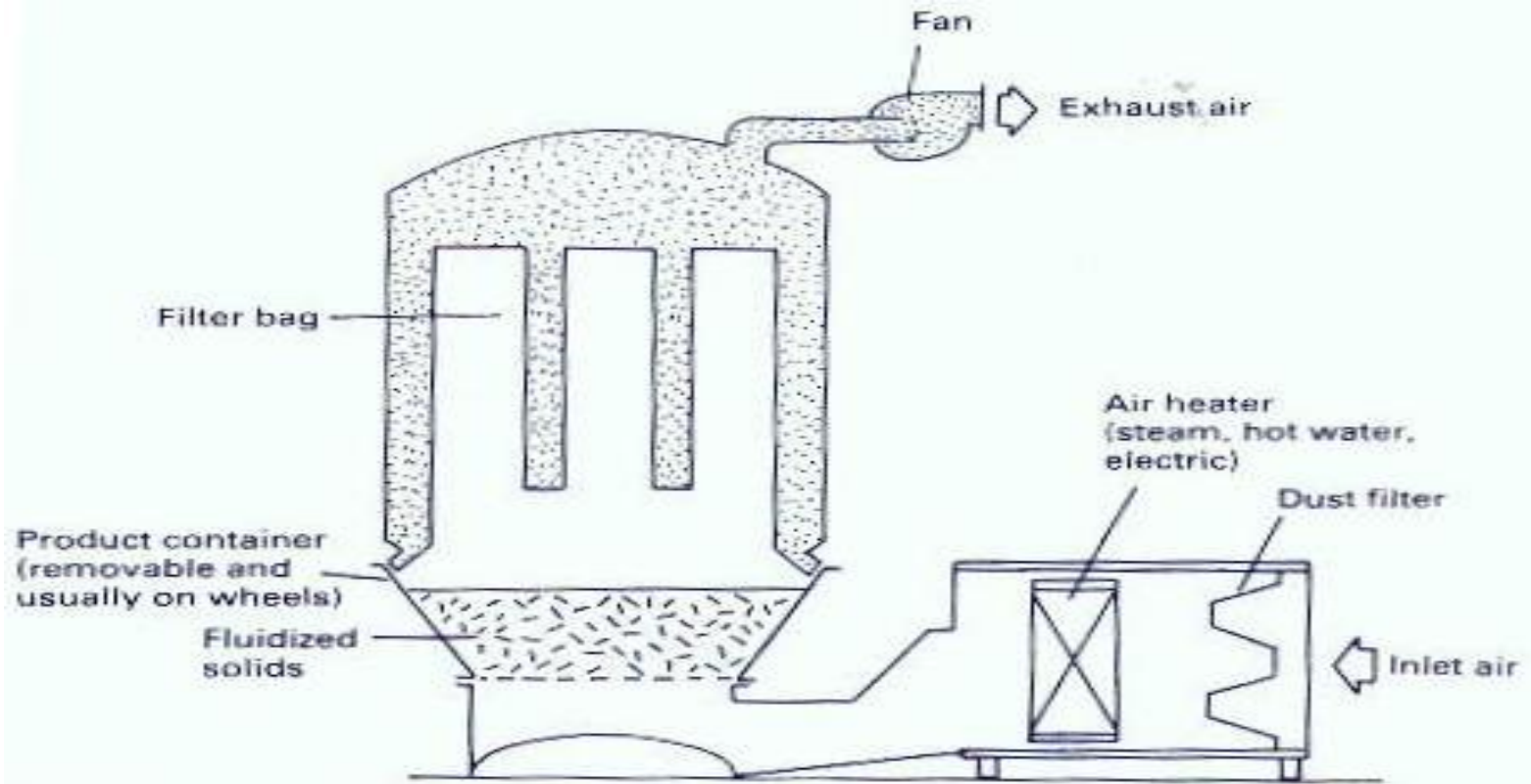
Rotary dryer.

Rotary Dryer...

- ❖ Heating is done either directly by **hot air current** or indirectly through an **outer shell containing steam tubes**
- ❖ The particles are passed through a rotating cylinder, counter-current to a stream of heated air
- ❖ Due to rotation of the cylinder, the material is turned over and drying takes place from *individual particles*
- ❖ *It is used for continuous drying on a large scale of any powdered or granular solids (up to 4 Mg/h)*

Fluidized Bed Dryer

- ❖ It is a stainless steel chamber with a perforated bottom into which the wet material to be dried is placed



Fluidized Bed Dryer...

- ❖ A gas is allowed to flow upward through a bed of particulate solids and the solids will be blown up and become partially suspended in the gas stream
- ❖ Up to 5 - 200 kg of the material can be dried in about 20-40 minutes

❖ **Advantages**

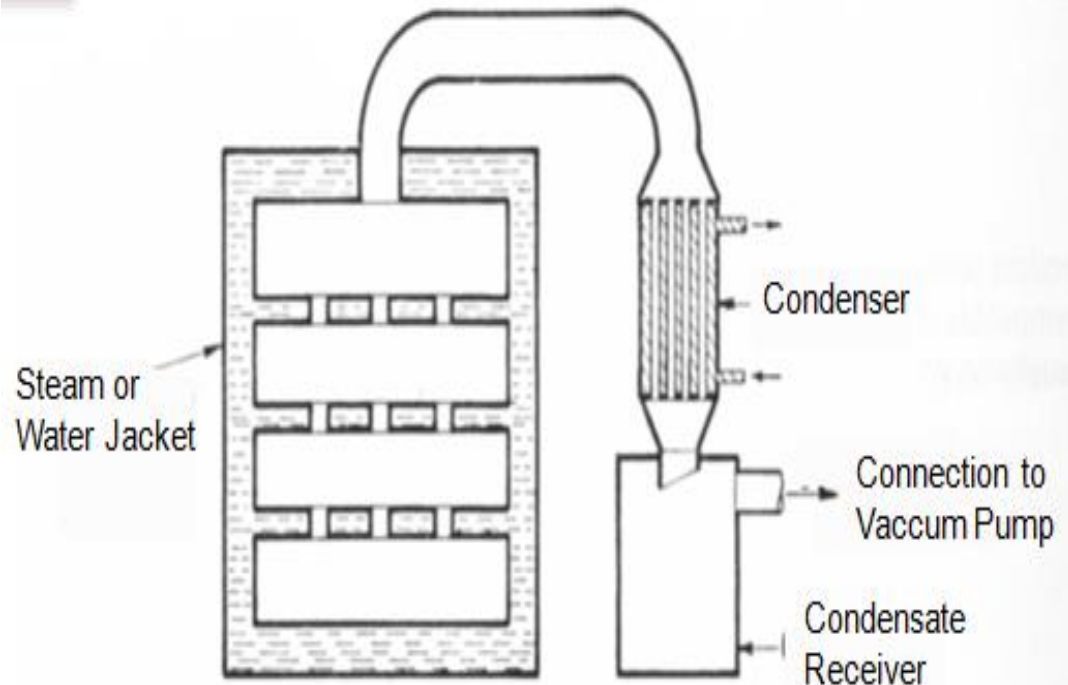
- ✓ Efficient heat exchange takes place
- ✓ Short drying time

❖ **Disadvantages**

- ✓ Larger quantities of fines are produced
- ✓ Too much wet granules may stick together

Vacuum Dryer

- ❖ The vacuum oven consists of a jacketed vessel to withstand vacuum within the oven.
- ❖ The oven is connected through a condenser and liquid receiver to a vacuum pump.
- ❖ Materials for drying can be spread on trays
- ❖ *Suitable for thermolabile materials*



Freeze Dryer

- ❖ Freeze drying is a process in which water is removed from liquid products by sublimation
- ❖ Freeze drying is a process used to dry extremely heat-sensitive materials.
- ❖ The material to be dried is first frozen and then subjected under a high vacuum to heat so that the frozen liquid sublimates leaving only the solid dried component of the original liquid
- ❖ Used for drying of *antibiotics, vaccines, blood products, enzymes etc*

Freeze Dryer...

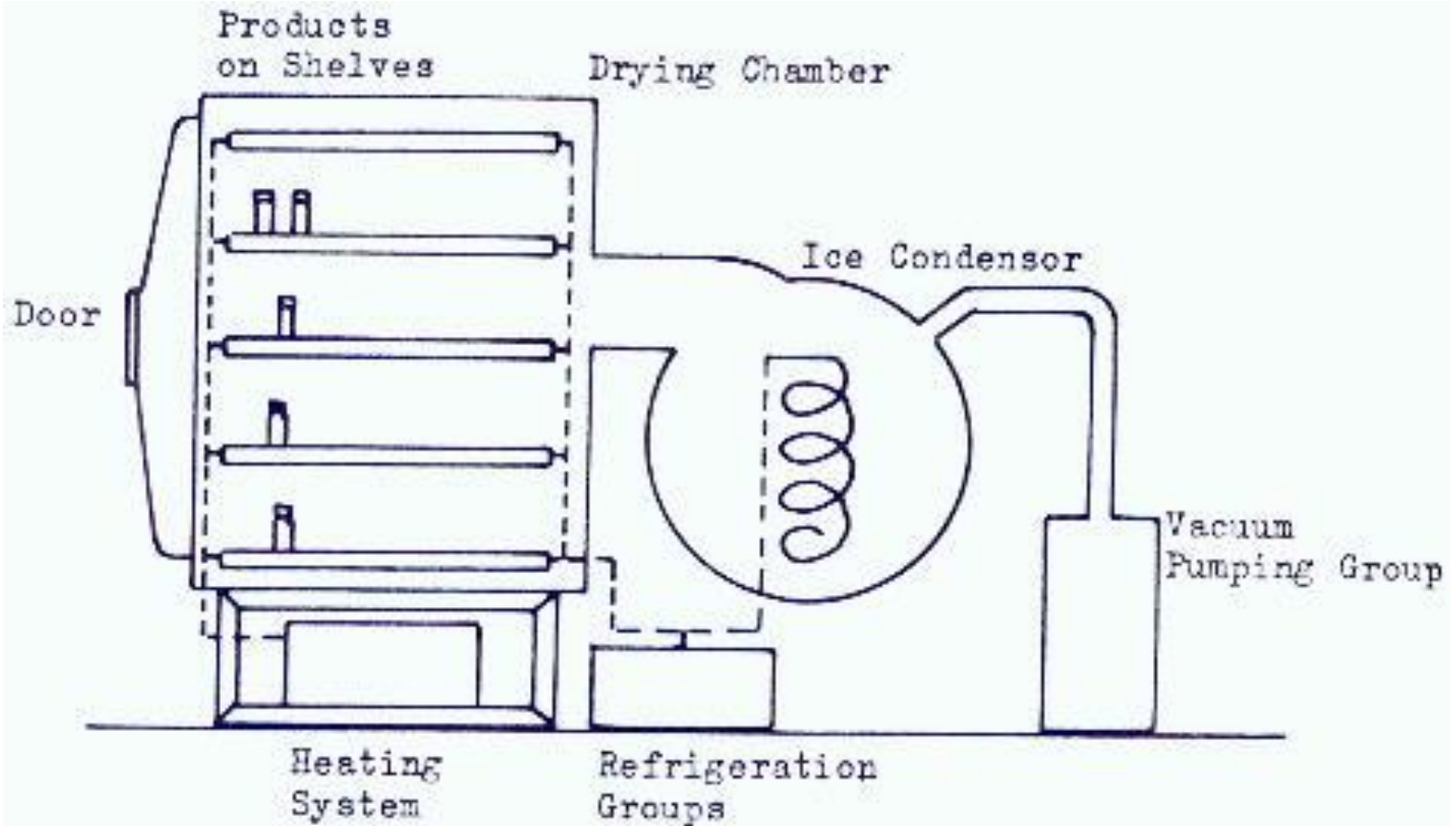


Fig: Freeze dryer

Applications of Drying

- ❖ For preparations of granules
- ❖ In the processing of materials e.g. $\text{Al}(\text{OH})_3$, spray dried lactose.
- ❖ To reduce bulk of materials thereby lowering cost of transportation and storage
- ❖ Preservation
- ❖ Increase stability of drugs
- ❖ Improve flowing property